

**Question:**

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding? Answer Choices: A. Radial artery B. Ulnar artery C. Princeps pollicis artery D. Brachial artery

**Reasoning + Answer:**

<think> The assistant is thinking about the fact that the anatomical snuffbox is a region on the wrist that contains a variety of structures, including blood vessels, tendons, and nerves. The region is also known for its thin skin, which can make it more susceptible to injury. The penetrating stab wound is likely to have caused significant damage to the blood vessels in the region, leading to excessive bleeding. The vascular surgeon is likely to be thinking about the fact that the anatomical snuffbox contains a variety of blood vessels, including the radial artery, which is a common cause of bleeding in this region. The surgeon may also be thinking about the fact that the bleeding may be caused by a variety of factors, such as a puncture wound, a laceration, or a crush injury. </think> <answer> A . Radial artery </answer>

